

Transcriptomic Reversal Screen — SSc Skin scRNA-seq

Generated: 2025-09-22 21:13 | Repo: scleroderma-scvi

Executive Summary

- Atlas-level reversal against LINCS L1000 perturbations across annotated cell types.
- Composite ranking emphasizes total reversal, fibroblast/myofibroblast selectivity, and significance.
- Broad cytotoxics deprioritized; MEK/ERK, EGFR axis, and BET families recur across clusters.
- All results are hypothesis-generating — not clinical guidance.

Shortlist (summary)

base_compound	moa	status	priority_final	total_score	max_rev	neglog10_best_p
PD-0325901	MEK inhibitor	Clinical-stage	3.22521119170242545	78637222358.08	98617820083.08	986178200832
gefitinib	EGFR inhibitor	FDA-approved	2.532581630368546.53	31205152123.44	27107125329.42	271071253192
WZ-3105	Unknown/other	Unknown	0.766406616825895.10	0374084988.25	15668421253.52	156684212389
PD-184352	MEK inhibitor	Clinical-stage	0.55392161470736.78	108550354378.29	44122406284.29	441224062844
BI-2536	PLK1 inhibitor	Clinical-stage	0.3394484653150285.22	73819132020.43	54353194569.35	543531945693
BET bromodomain inhibitor	BET bromodomain inhibitor	Unknown	0.275805214587428.08	66661980768.28	87108855889.28	871088558894
JNK-9L	JNK inhibitor	Unknown	0.09709532175928.22	2760278725.15	31801773876.95	318017733769
BET bromodomain inhibitor	BET bromodomain inhibitor	Clinical-stage	0.161889760962277.73	8993348708.35	54265116359.55	542651163592
sirolimus	mTOR inhibitor	FDA-approved	0.174531468283768.08	79103988318.36	35583145518.36	355831455193
pelitinib	EGFR inhibitor	Unknown	0.437989162439029.55	44194868078.09	38164055118.09	3816405511827
GSK-429286A	Unknown/other	Unknown	0.466606670117835.22	89227367058.48	87108855889.28	871088558894
dovitinib	Unknown/other	Unknown	0.661732736803342.19	0983007719.01	38853247702.44	3885324770244
olaparib	Unknown/other	Unknown	0.003007155260799.40	1742063222.50	2690219217.53	269021921713
KIN001-043	Unknown/other	Unknown	1.32517203629619.79	0147319118.54	2976594140.89	297659414286
withaferin-a	Unknown/other	Unknown	1.99457725293878.73	0771502790.25	5778833890.15	577883389414

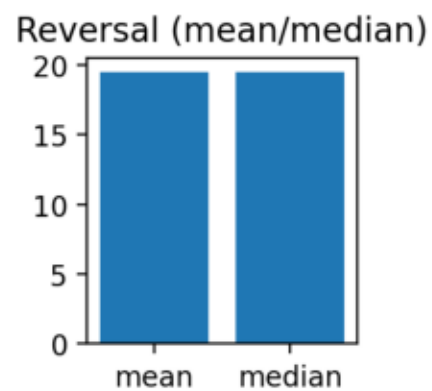
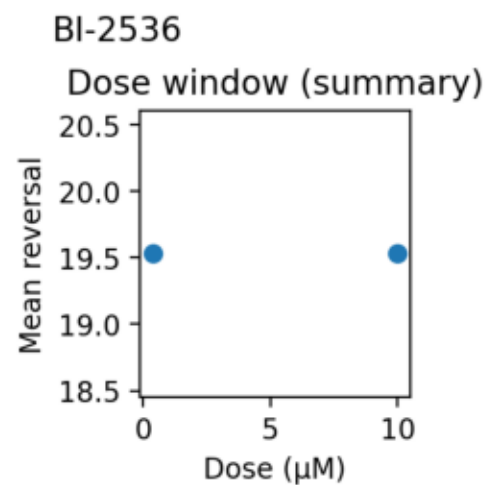
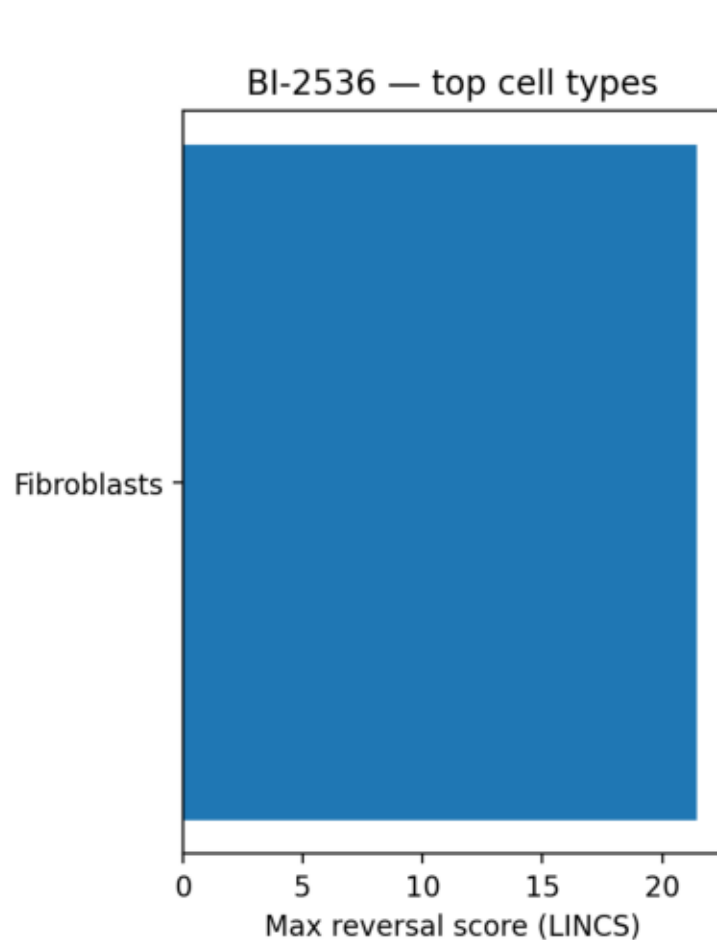
Dose/Time Consistency (summary)

base_compound	n_profiles	times_seen	dose_min	dose_max	score_mean	score_sd	best_p_min
PD-184352	1	24.0	10.0	22.0944	1224062844	5.07677316	7564584e-23
BI-2536	2	24.0	0.37	19.5087	485162699462	3.65970435	8248452e-22
sirolimus	1	24.0	10.0	18.0635	5831455193	3.439075303	6555272e-19
PD-0325901	6	3.0,24.0	1.11	18.30249	10495975854	69921396014	7706718e-29
GSK-429286A	1	24.0	0.04	18.2387	1088558894	5.143858607	8551735e-19
I-BET151	4	24.0	1.11	17.92193	95686765885	5.0204629538	6770615e-19
gefitinib	8	3.0,24.0	0.12	16.0402	9746952662	26582833620	1993481e-24
I-BET	2	24.0	3.33	14.26170	4047546751708	1099252034	7057674e-19
WZ-3105	14	24.0	0.04	14.10176	973040634899	3.8226081023	7445607e-24
pelitinib	3	3.0,24.0	3.33	13.06664	886334358942	29082909398	0412015e-18
dovitinib	2	24.0	3.33	11.36789	836193165613	78134900715	3729023e-14
JNK-9L	6	24.0	0.37	10.96594	56968454865	71918342508	4616633e-16
olaparib	3	24.0	0.04	10.37851	022502860499	82834275984	2132749e-12
withaferin-a	3	24.0	1.11	9.70643	484533878906	382348908109	2565754e-13
KIN001-043	7	3.0,24.0	1.11	8.80740	067682030631	59873738420	1118416e-14

LINCS Coverage (cell types per compound)

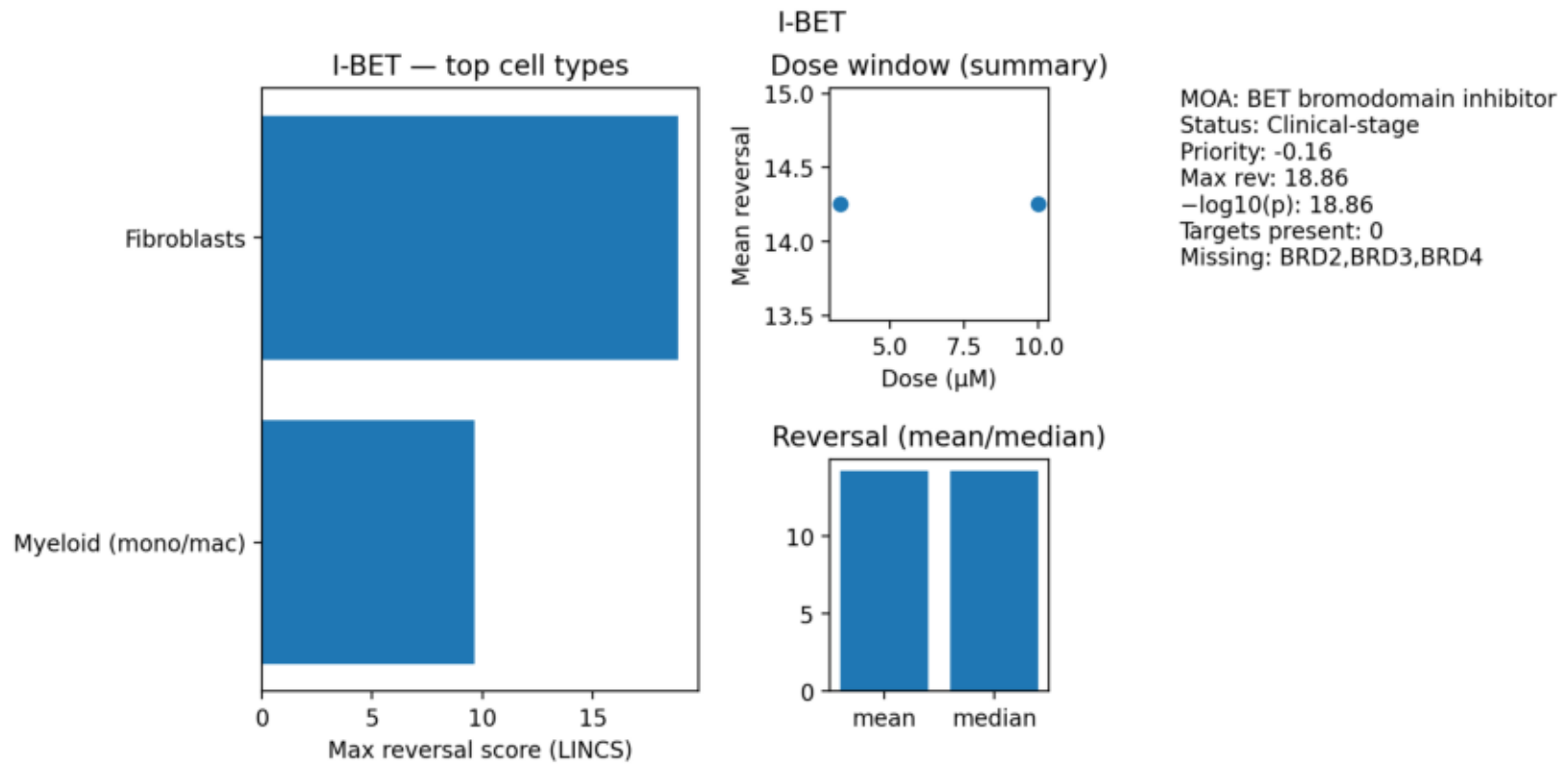
compound	n_celltypes
LJP006 HA1E 24H-CGP-60474-0.37	5
LJP006 HA1E 24H-alvocidib-10	4
LJP006 HA1E 24H-mitoxantrone-3.33	4
LJP006 HME1 24H-gefitinib-0.37	4
LJP006 MCF10A 24H-WZ-4-145-3.33	4
LJP006 HA1E 24H-CGP-60474-3.33	3
LJP006 HA1E 24H-CGP-60474-10	3
LJP006 HA1E 24H-alvocidib-3.33	3
LJP006 HA1E 24H-CGP-60474-1.11	3
LJP006 HCC515 24H-CGP-60474-10	3
LJP006 HME1 24H-AS-605240-3.33	3
LJP006 HME1 24H-PD-0325901-1.11	3
LJP005 A375 24H-PLX-4720-0.04	3
LJP006 HA1E 24H-AZD-5438-10	3
LJP006 A375 24H-alvocidib-10	2
LJP006 HCC515 24H-AT-7519-3.33	2
LJP006 HA1E 24H-alvocidib-0.37	2
LJP005 A375 24H-CGP-60474-1.11	2
LJP005 MDAMB231 24H-WZ-3105-3.33	2
LJP006 A375 24H-JNK-9L-10	2
LJP005 MCF10A 24H-PF-431396-3.33	2
LJP006 A549 24H-mitoxantrone-10	2
LJP006 MDAMB231 24H-CGP-60474-10	2
LJP006 MDAMB231 24H-CGP-60474-0.04	2
LJP006 PC3 24H-CGP-60474-10	2

One-pager — BI-2536

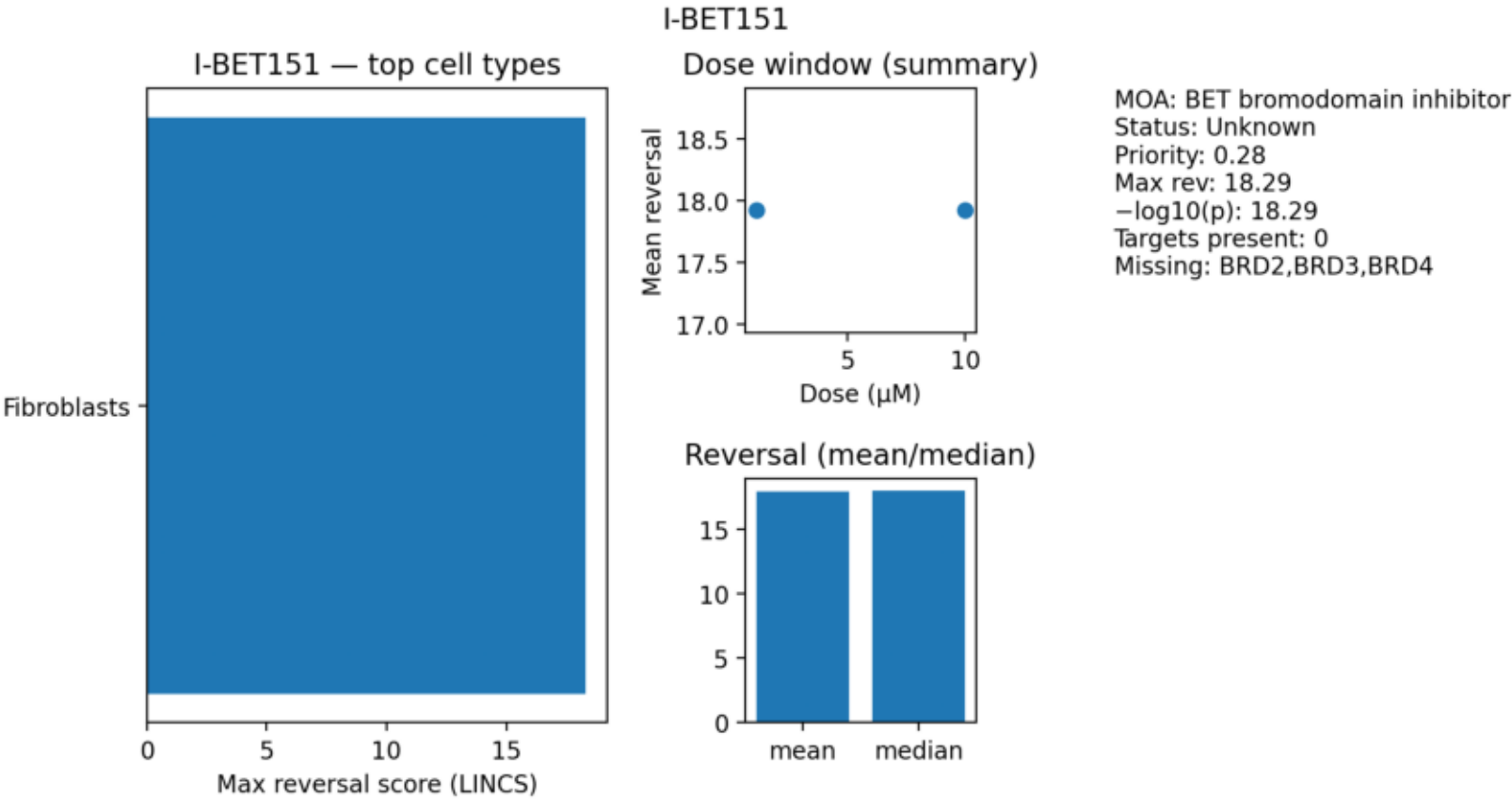


MOA: PLK1 inhibitor
Status: Clinical-stage
Priority: 0.38
Max rev: 21.44
 $-\log_{10}(p)$: 21.44
Targets present: 1
Missing: nan

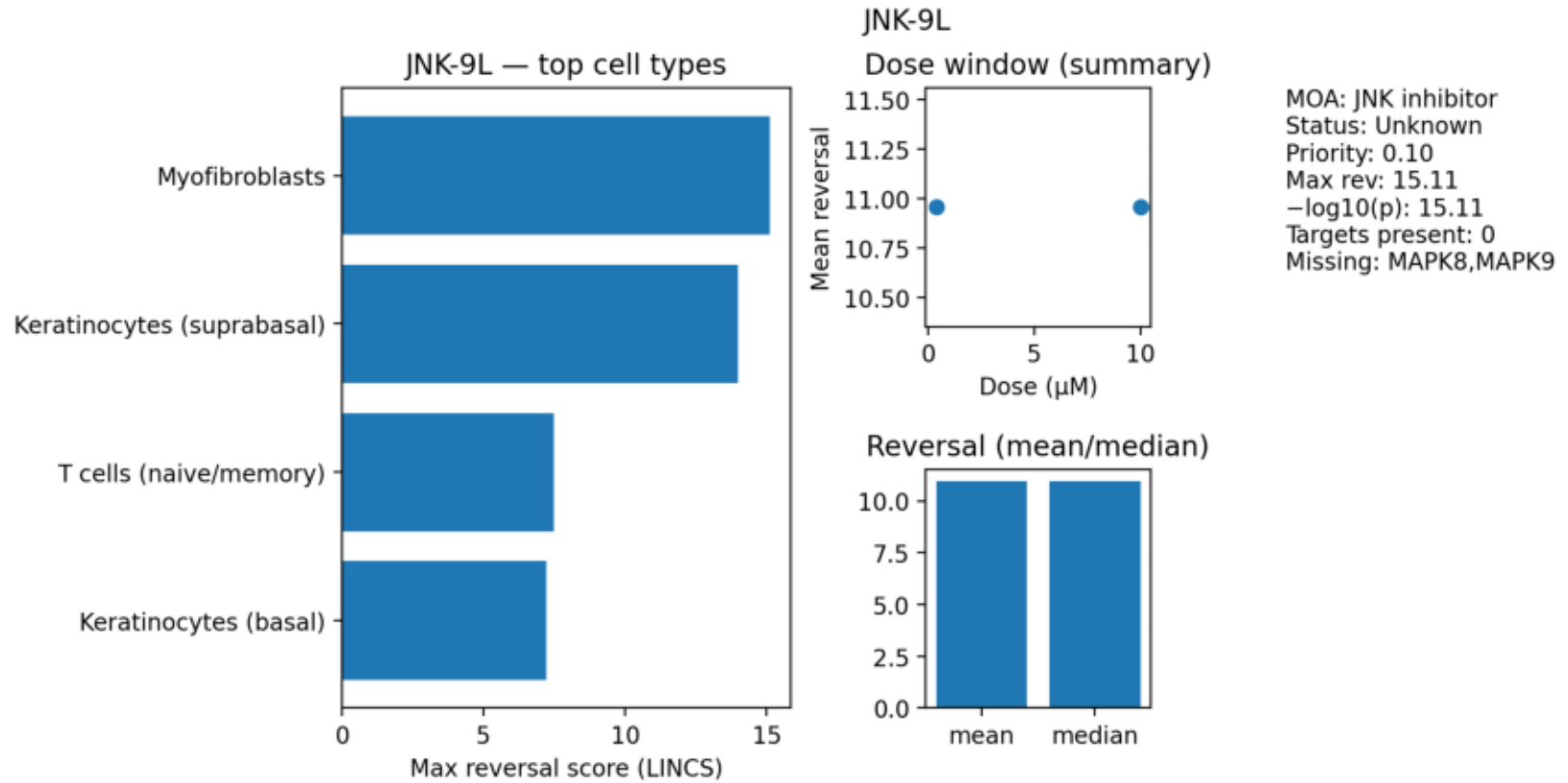
One-pager — I-BET



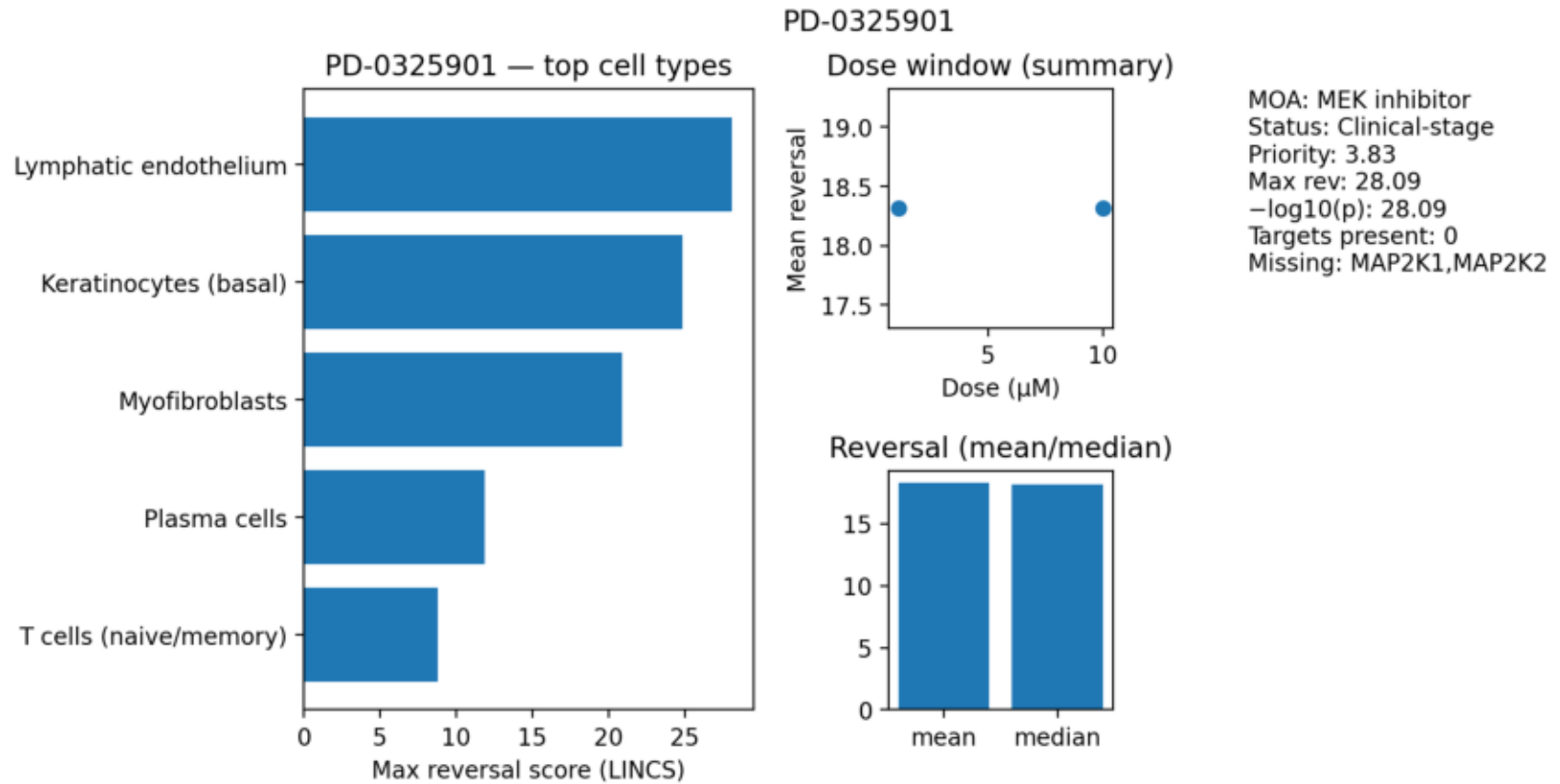
One-pager — I-BET151



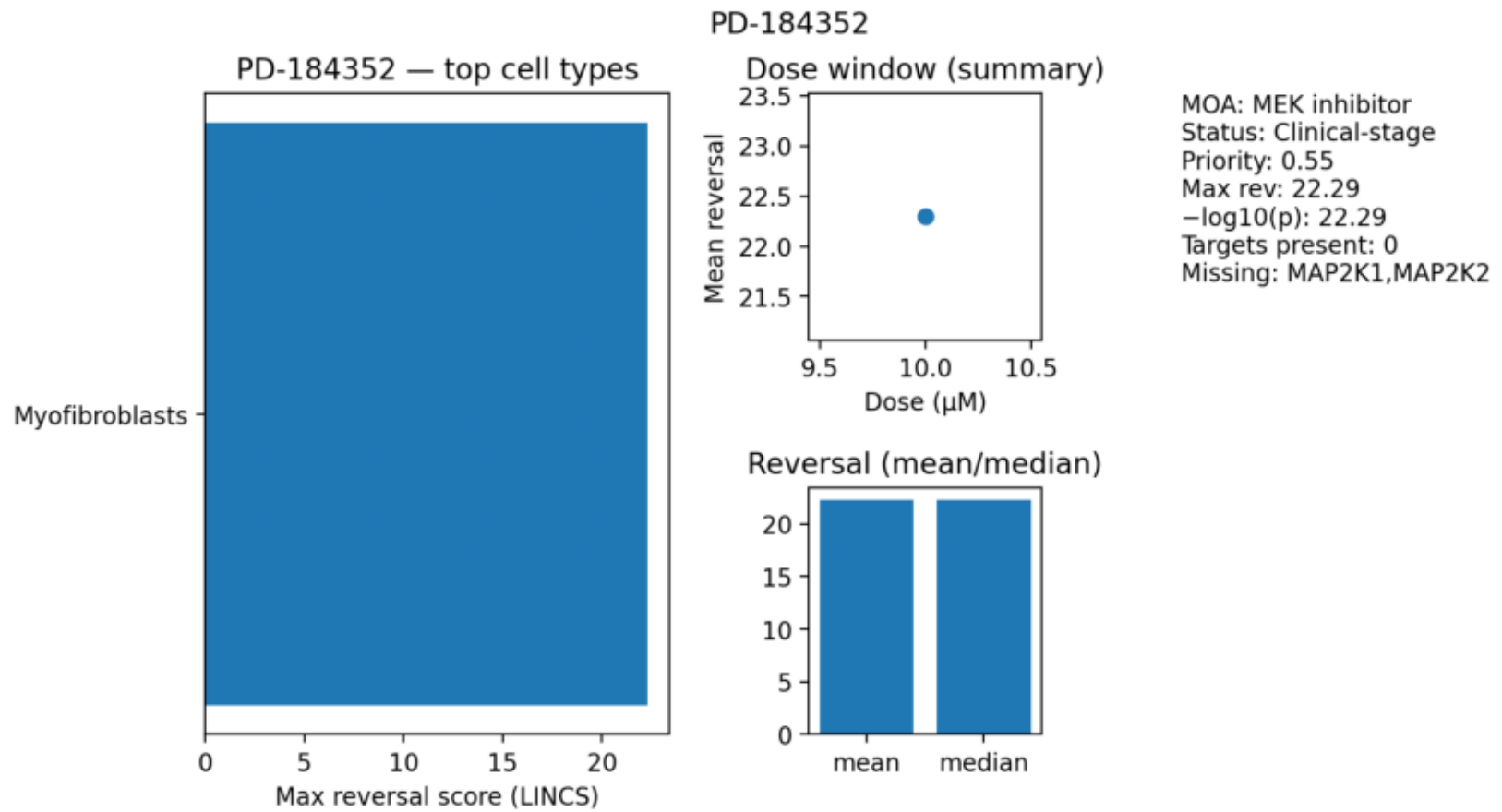
One-pager — JNK-9L



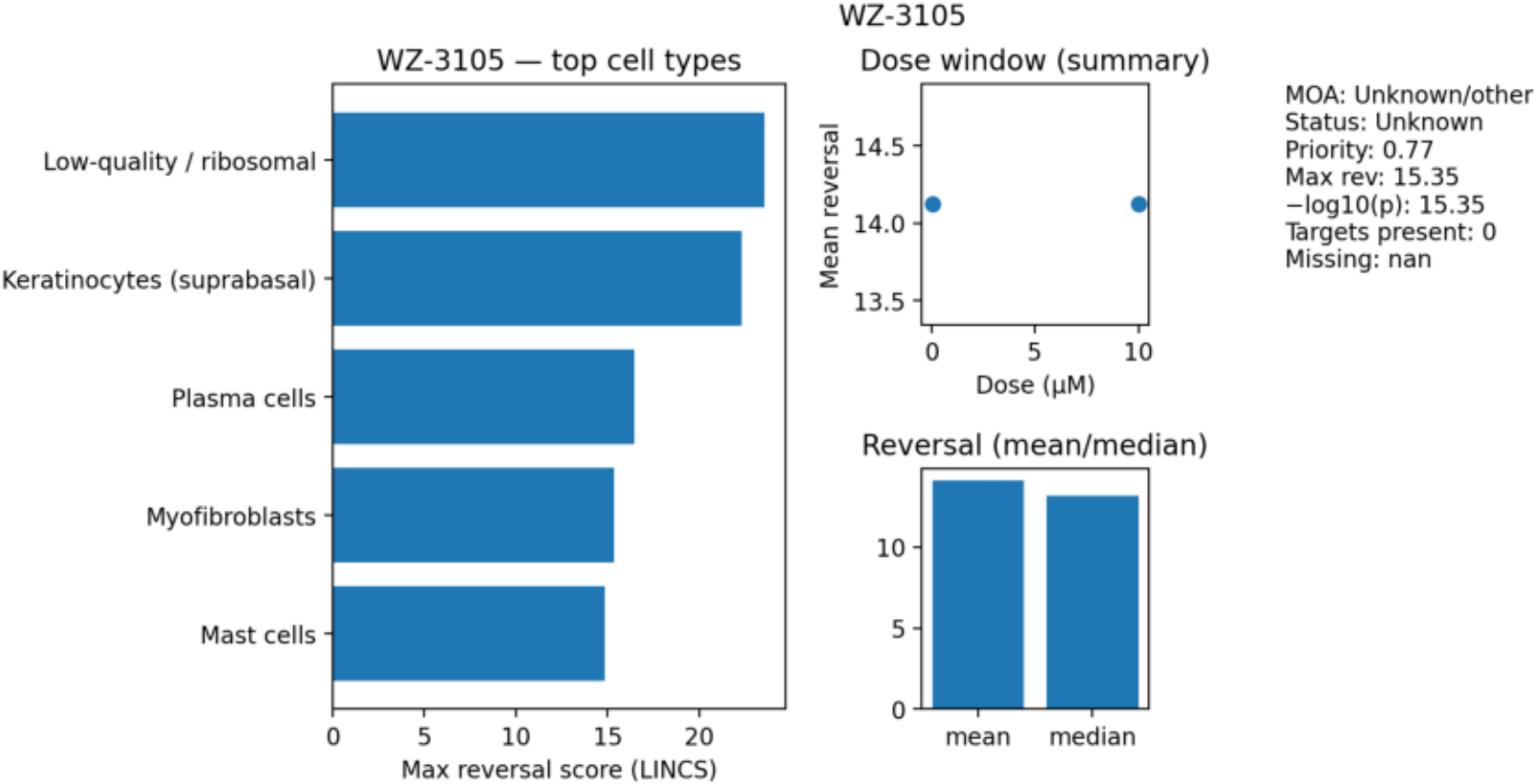
One-pager — PD-0325901



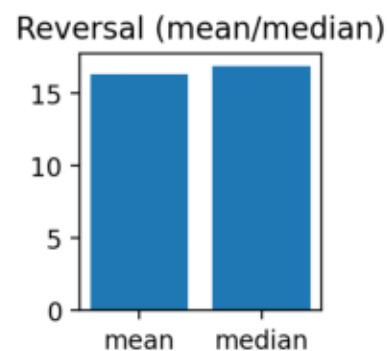
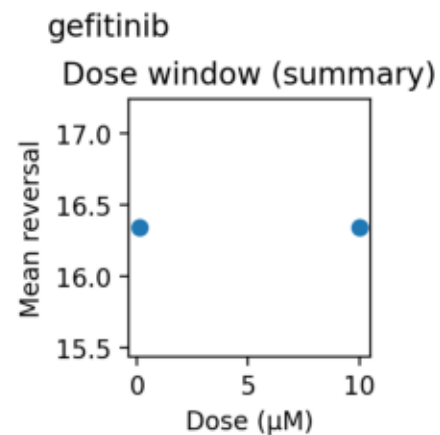
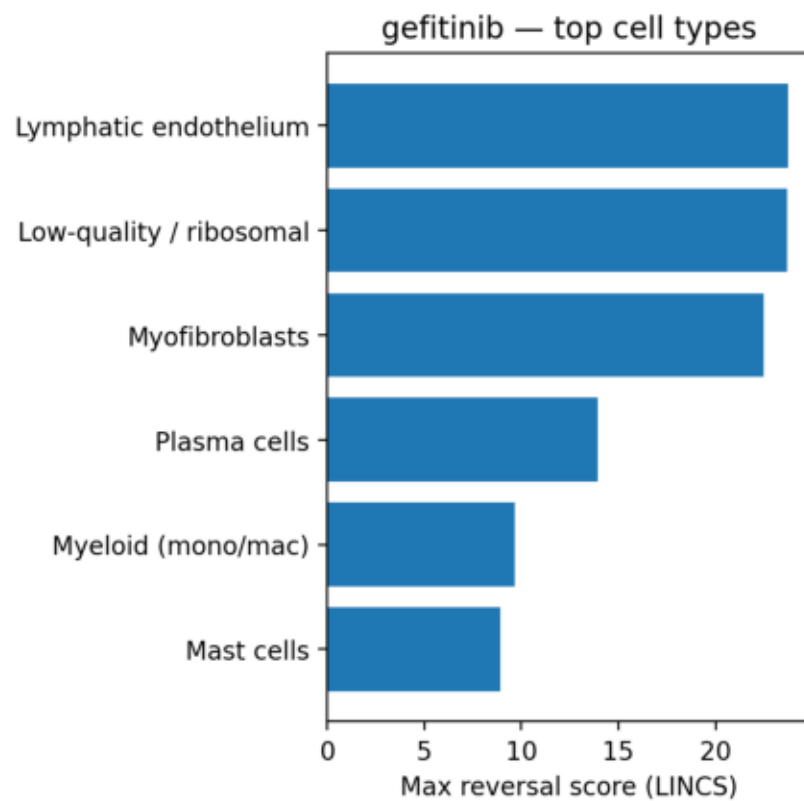
One-pager — PD-184352



One-pager — WZ-3105

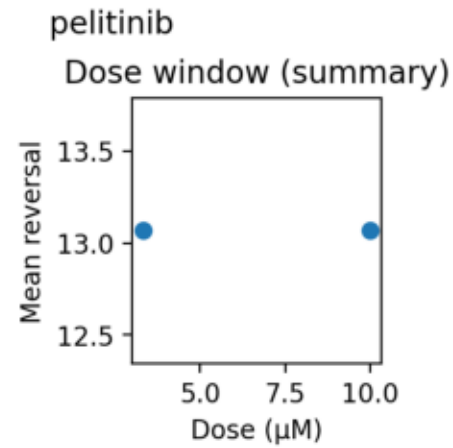
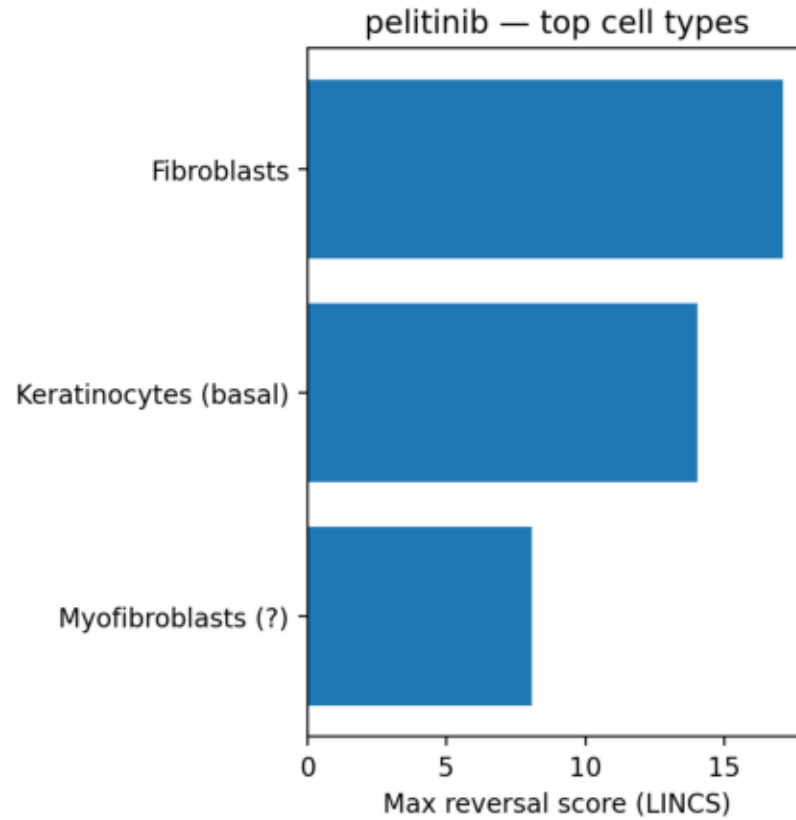


One-pager — gefitinib

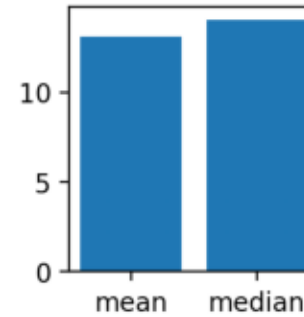


MOA: EGFR inhibitor
Status: FDA-approved
Priority: 2.53
Max rev: 23.74
-log10(p): 23.74
Targets present: 0
Missing: EGFR

One-pager — pelitinib



Reversal (mean/median)



MOA: EGFR inhibitor
Status: Unknown
Priority: -0.44
Max rev: 17.09
-log10(p): 17.09
Targets present: 0
Missing: EGFR

One-pager — sirolimus

